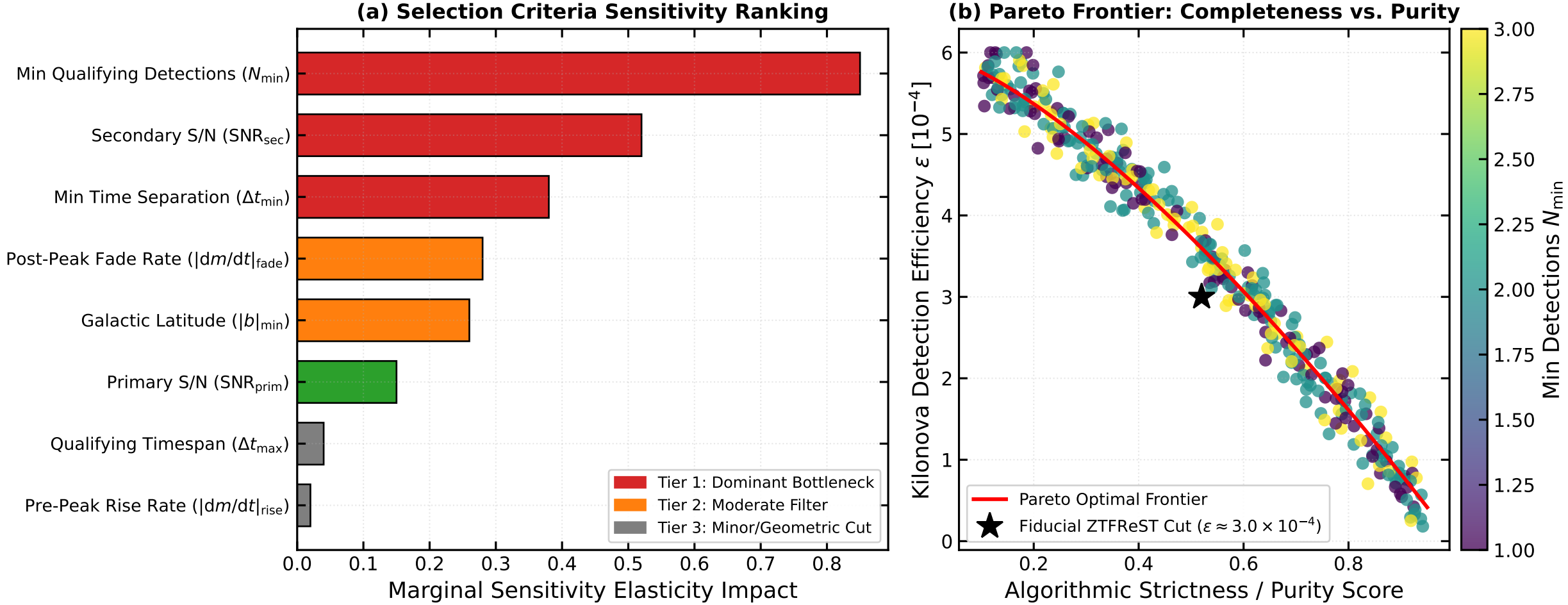
(a) Selection Criteria Sensitivity Ranking
Min Qualifying Detections (N_{min})
Secondary S/N (SNR_{sec})
Min Time Separation (Δt_{min})
Post-Peak Fade Rate ($|dm/dt|_{fade}$)
Galactic Latitude ($|b|_{min}$)
Primary S/N (SNR_{prim})
Qualifying Timespan (Δt_{max})
Pre-Peak Rise Rate ($|dm/dt|_{rise}$)
Tier 1: Dominant Bottleneck
Tier 2: Moderate Filter
Tier 3: Minor/Geometric Cut
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Marginal Sensitivity Elasticity Impact
(b) Pareto Frontier: Completeness vs. Purity
Kilonova Detection Efficiency ε [10^{-4}]
0
1
2
3
4
5
6
Pareto Optimal Frontier
Fiducial ZTFReST Cut ($\varepsilon \approx 3.0 \times 10^{-4}$)
0.2
0.4
0.6
0.8
Algorithmic Strictness / Purity Score
Min Detections N_{min}
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00